

北京少年人工智能学院 (海淀)
BEIJING YOUTH AI
ACADEMY (HAIDIAN)

Large Language Models for Classical Chinese Poetry Appreciation

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Instructed by Gao Yan

(The English version only includes the technical report section)

About Our Team



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Project Background (Motivation)

- **Core Value:** Irreplaceable for cultural heritage, language accumulation, and aesthetic education.
- **The Problem:** Recitation $\neq$ Comprehension. A true connection hasn't been established.

Dilemma	Student's Voice
Memorize vs. Comprehend	"Why does this line represent responsibility?"
Memorize vs. Create	I can recite it, but how do I write my own?
Create vs. Improve	I wrote a poem, but how do I make it better?

Our Solution: Echo

- What is Echo?
 - Echo means **reflection and resonance**.
 - Our model is based on **the rhythm and aesthetics** of classical poetry, serving as a connoisseur and mentor for middle school students learning classical poetry: understanding not only the rhythm of poetry but also the needs of students, responding to and assisting every learner.



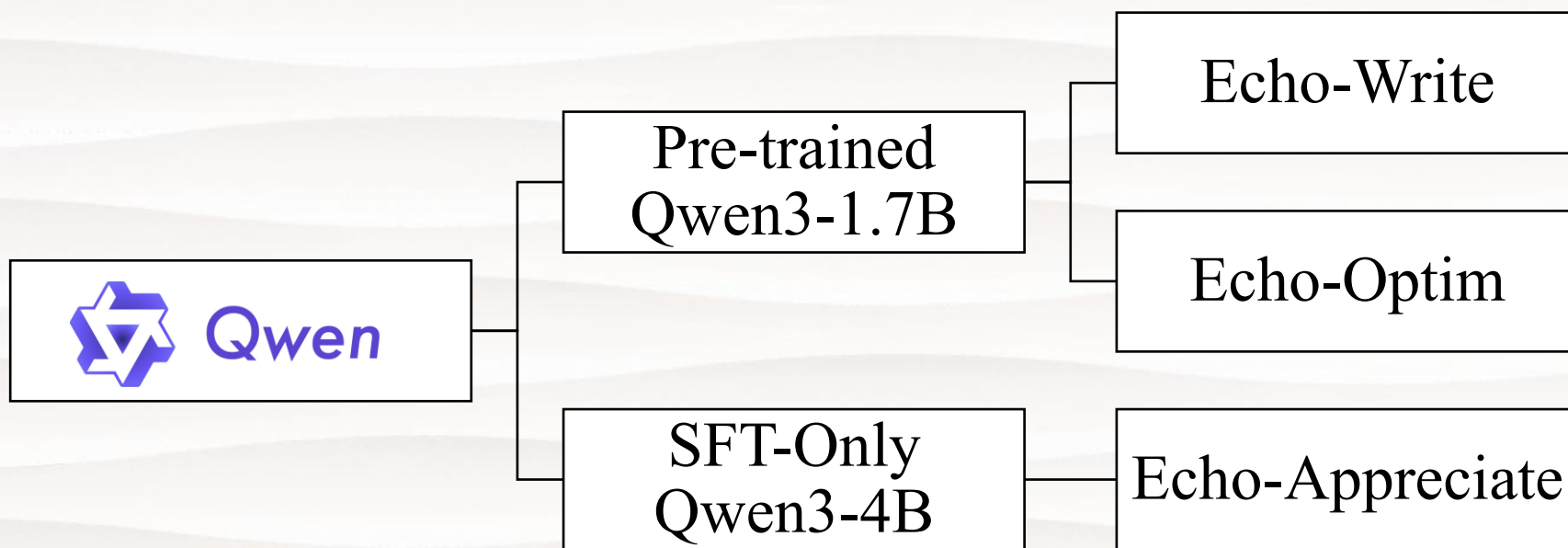
Technical Report

Problems, Solutions and Evaluation

Problems and Solutions

Problems	Solution
Data Scarcity: How to improve training quality?	Synthesize brand-new poetry and analysis via specialized methods.
Low Efficiency: 3 models sharing 1 dataset—how to optimize?	Build a unified data pipeline to standardize data processing.
Incremental Pre-training: Is it necessary?	Testing showed it caused unexpected <i>catastrophic forgetting</i> in the analysis model.
Evaluation: How to evaluate 3 models cost-effectively?	LLM-based grading + Human questionnaires.

Base Model Info



Data Synthesis and Model Training Process

1. Structed Requirements

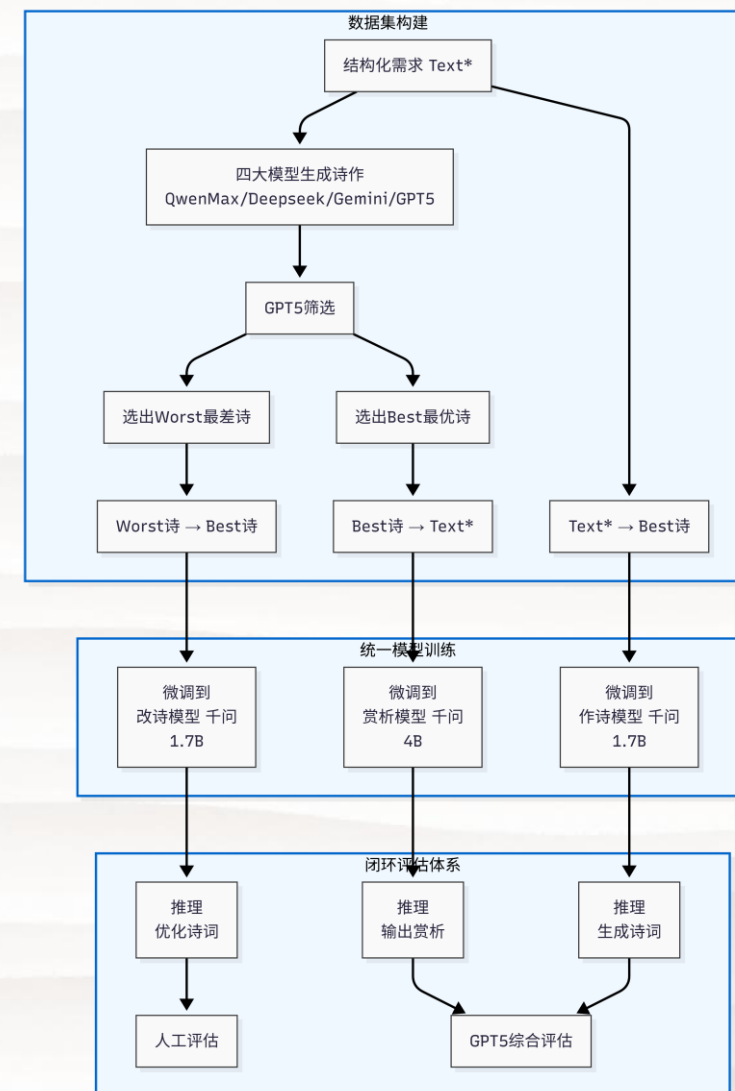
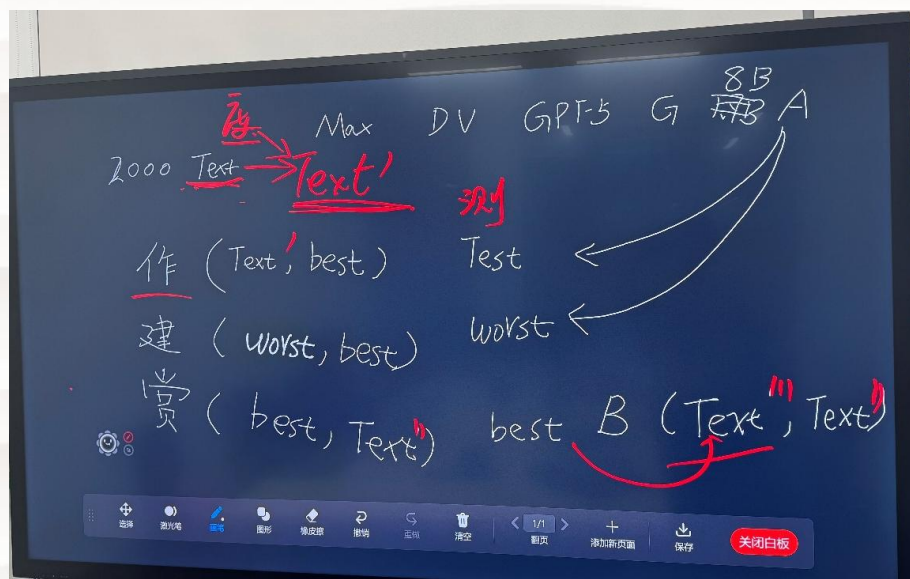
- Core: Parses user prompt → Standardizes system input.

2. LLM Generation & Filtering

- Core: Multi-LLM generation → GPT-5 filters Best / Worst.

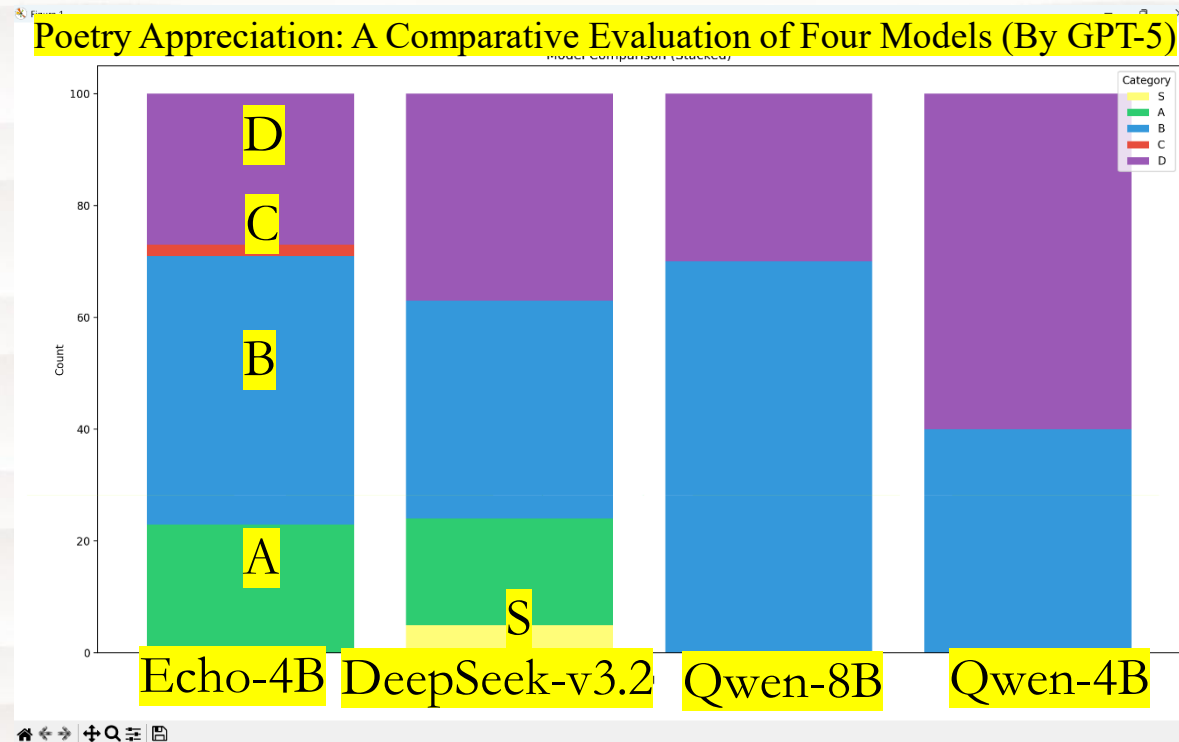
3. Small Model Fine-Tuning

- Core: Qwen3-1.7B + 300K domains pre-training.
- Outputs: 3 Domain-Specific Models: Generation / Revision / Appreciation.



Evaluate Result I: Appreciating

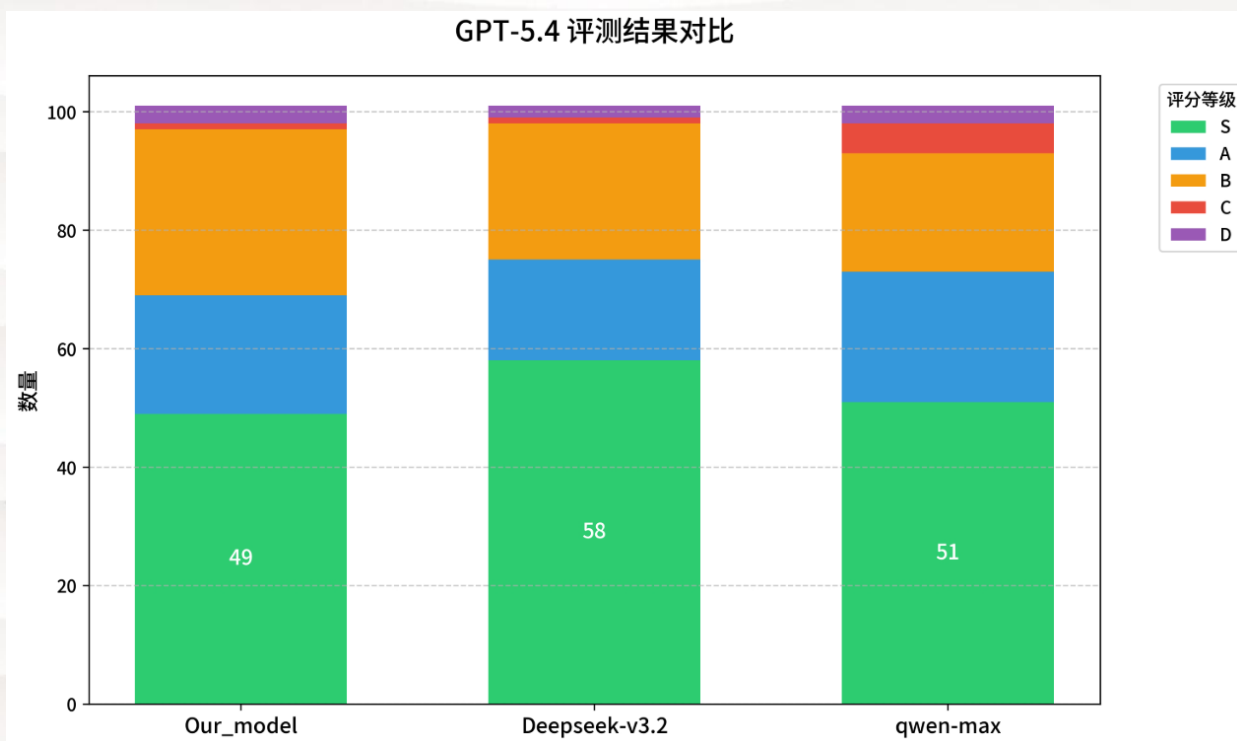
By analyzing 100 questions and applying the GPT-5 scoring (SABCD) method, we obtained the following data:



- Top-Tier Performance(S/A Grade: ~30%)
- Industry-Leading Stability(D Grade: ~30%)
- Highly Effective Fine-Tuning

Evaluate Result II: Writing

Based on the above evaluation, the following classifications are obtained after scoring by GPT-5:



Small Model, Giant Performance: Beat models with **100x+** of parameters.
Unmatched Stability: C/D Grade (Failure Rate): Only **4%**.

Evaluate Result III: Polishing

Because the polishing comparison evaluation is highly subjective, the AI model performed poorly in this task, so we used a questionnaire to conduct the evaluation.

* 1. 1. 请按您认为的五首诗词好坏程度排序【请选择全部项并排序】

- ☐ 大漠孤烟卷暮尘，长河落日照征人。弓如霹雳的卢奔。意气凌云催战鼓，挥鞭直指虏营门。方道谈笑定乾坤。
- ☐ 《大漠征歌》大漠孤烟卷暮尘，长河落日照征人。弓如霹雳龙腾跃，马似银鳞电闪巡。意气凌云催战鼓，挥鞭直指虏营门。方道谈笑定乾坤，一箭惊飞万里云。
- ☐ 大漠孤烟起暮愁，长河落日映征裘。弓如霹雳射星流。万里风尘平虏阵，一骑天边报凯讴。男儿志在斩匈奴。
- ☐ 大漠孤烟凝暮霭，长河落日映征魂。弓鸣霹雳惊卢越。意气凌霄催战角，挥鞭遥指虏营分。谈笑方道定乾坤。
- ☐ 大漠孤烟卷暮尘，长河落日没征轮。弓刀霹雳的卢奔。意气干云催画角，挥鞭直指玉关尘。笑谈天地定三分。

Initial goal: To establish a preliminary understanding of the overall effect through multiple version polling.

Problems: Significant version redundancy makes it difficult for users to distinguish subtle differences, and data accuracy is limited.

Iteration: Simplify and focus on core version battles.

Results: By comparing models pairwise, accurately capture user preferences, making evaluation data more precise and convincing.

* 1. 请对比下面两首古诗润色版本，综合判断哪一版更符合古诗审美与表达习惯？

- ☐ 大漠孤烟起暮愁，长河落日映征裘。弓如霹雳射星流。万里风尘平虏阵，一骑天边报凯讴。男儿志在斩匈奴。
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Evaluate Result III: Polishing

We received over 1200 annotations.

对比组合	题目数	我方胜	对方胜	平局	我方票数	对方票数	得票率
VS 原文 (T0)	50	26	19	5	170	139	55.0%
VS baseline	50	27	18	5	172	134	56.2%
VS deepseek	50	27	18	5	176	137	56.2%
VS qwen-max	50	24	22	4	171	137	55.5%
总计	200	104	77	19	689	547	55.8%

1. Global & Absolute Leadership

- **Dominant Win Rate:** **55.8%** total votes (104 Wins, 77 Losses, 19 Ties).
- **Positive Gains:** Achieved positive returns across all comparison groups.

2. Empowering Student Writing

- **Outperforming Originals:** Secured a **55.0%** vote rate over students' original texts.
- **Solving the Pain Point:** Effectively addresses the dilemma of *"Able to write, unable to revise."*

3. Small Model, Giant Performance

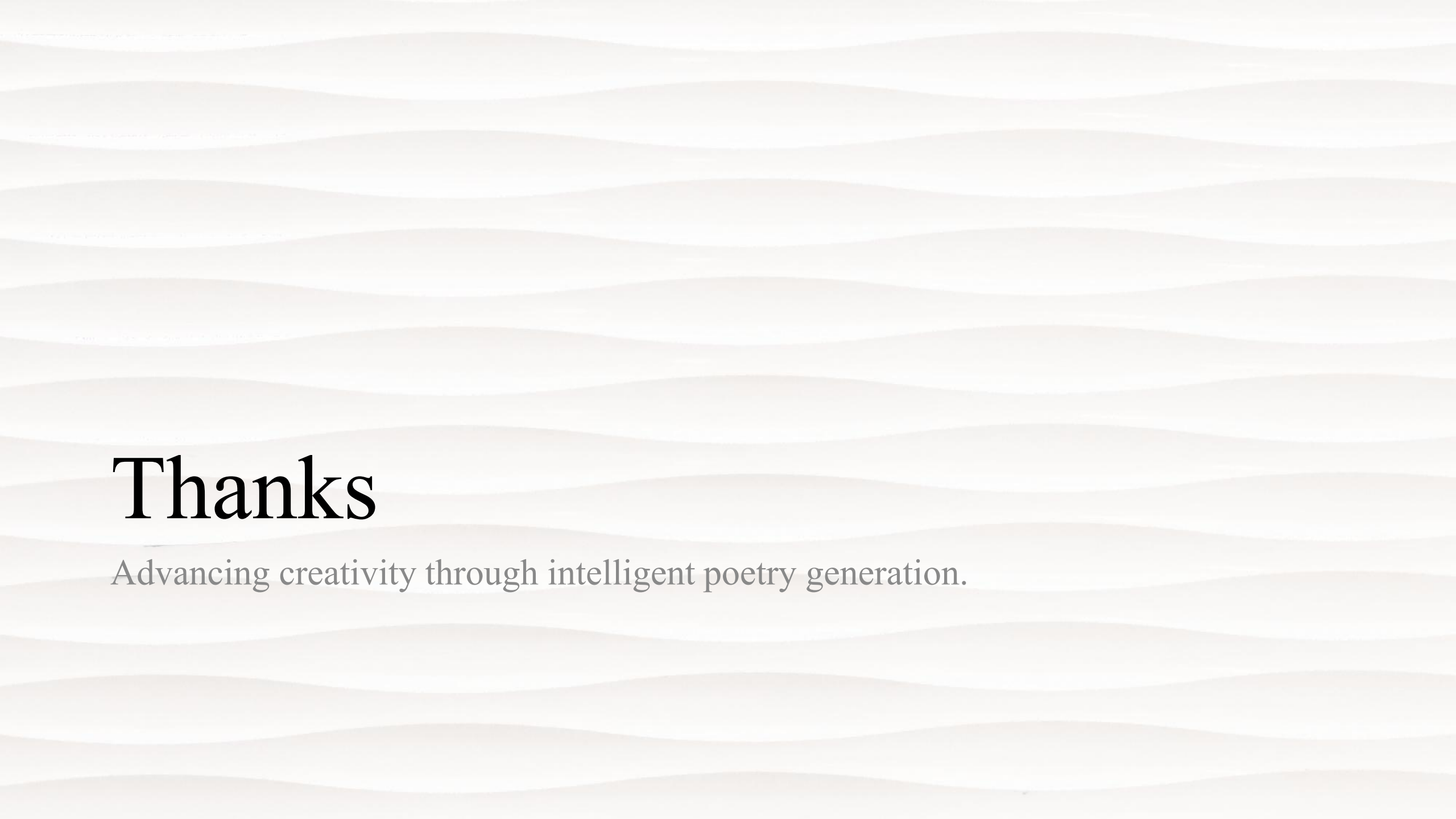
- **Beating the Giants:** Achieved **56.2%** and **55.5%** vote rates against tier-1 LLMs.
- **Top-Tier Rivals:** Outperformed industry leaders including DeepSeek and Qwen-MAX.

4. Highly Effective Fine-Tuning

- **Crushing the Baseline:** Substantially outperforms the native baseline model.
- **Proven Strategy:** Validates the absolute effectiveness of our domain-specific vertical fine-tuning.

Some Findings

- **1. Structural & Linguistic Focus**
 - Bypassed rote memorization; concentrated entirely on word choice and syntactic arrangement.
- **2. Strategic Component Pruning**
 - Stripped out background and translation to fully prioritize core poetry appreciation.
- **3. Zero-Gap Model Robustness**
 - Maximized stability to ensure identical performance on both new creations and timeless classics.



Thanks

Advancing creativity through intelligent poetry generation.